

MACHINE MONITORING AND DATA ANALYSIS USING NUMPY AND PANDAS

Aim: To write and execute a Python program for analyzing machine operating data using NumPy and Pandas.

Interactive Development Environment (IDE) Used : Jupyter Notebook

Programming Language Used : Python

Python Concepts used:

NumPy and Pandas are important Python libraries used for numerical computing and data analysis. NumPy provides functions for efficient numerical operations on arrays and numerical data. Pandas provides data structures and functions for organizing, analyzing, selecting and processing tabular data. A Pandas DataFrame is a two-dimensional table consisting of rows and columns. In addition, Python supports indexing, conditional expressions and functions for processing data.

Python Concept	Example	Description
NumPy	<code>np.mean()</code>	Calculates the arithmetic mean of numerical data.
Pandas	<code>pd.DataFrame()</code>	Creates a DataFrame from structured data.
DataFrame	<code>df</code>	Stores data in rows and columns.
Mean	<code>np.mean(x)</code>	Returns the average of the values.
Maximum	<code>x.max()</code>	Returns the largest value.
Index	<code>idxmax()</code>	Returns the index of the maximum value.
Condition	<code>x > 3.0</code>	Checks whether a condition is satisfied.
Column Selection	<code>df["Temperature_C"]</code>	Selects a column from a DataFrame.
Conditional Selection	<code>np.where()</code>	Selects values based on a condition.

Data Analysis Operation / Concept

Data Analysis Operation / Concept	Example Code	Description
Import NumPy	<code>import numpy as np</code>	Imports the NumPy library and assigns the alias np.
Import Pandas	<code>import pandas as pd</code>	Imports the Pandas library and assigns the alias pd.
Create DataFrame	<code>pd.DataFrame(data)</code>	Creates a DataFrame from structured data.
Select a Column	<code>df["Temperature_C"]</code>	Selects the specified column from a DataFrame.
Calculate Mean	<code>np.mean(df[...])</code>	Calculates the arithmetic mean of the selected numerical data.

Round a Value	<code>round(value, 2)</code>	Rounds a numerical value to two decimal places.
Find Maximum	<code>df["Vibration_mm_s"].max()</code>	Returns the maximum value in the selected column.
Find Maximum Index	<code>df["Vibration_mm_s"].idxmax()</code>	Returns the index of the row containing the maximum value.
Select Using loc	<code>df.loc[index, "Machine"]</code>	Selects data using row and column labels.
Conditional Operation	<code>np.where(condition, A, B)</code>	Selects A when the condition is true and B otherwise.
Select Multiple Columns	<code>df[["Machine", "Condition"]]</code>	Selects the specified columns from the DataFrame.

Python Program:

```
import numpy as np
import pandas as pd

# Machine data
data = {
    "Machine": ["M1", "M2", "M3", "M4", "M5"],
    "Speed_RPM": [1200, 1500, 1800, 1400, 1600],
    "Temperature_C": [65, 72, 85, 68, 78],
    "Vibration_mm_s": [2.1, 2.8, 4.5, 2.4, 3.2],
    "Power_kW": [5.2, 6.5, 8.0, 5.8, 7.1]
}

# Create DataFrame
df = pd.DataFrame(data)

print("Machine Data:")
print(df)

# Calculate average values
average_temperature = np.mean(df["Temperature_C"])
average_vibration = np.mean(df["Vibration_mm_s"])
average_power = np.mean(df["Power_kW"])

print("\nAverage Temperature:",
      round(average_temperature, 2), "°C")

print("Average Vibration:",
      round(average_vibration, 2), "mm/s")

print("Average Power:",
```

```

round(average_power, 2), "kW")

# Find machine with maximum vibration
max_vibration = df["Vibration_mm_s"].max()

machine = df.loc[
    df["Vibration_mm_s"].idxmax(),
    "Machine"
]

print("\nMaximum Vibration:",
      max_vibration, "mm/s")

print("Machine with Maximum Vibration:", machine)

# Classify machine condition
df["Condition"] = np.where(
    df["Vibration_mm_s"] > 3.0,
    "Needs Inspection",
    "Normal"
)

print("\nMachine Condition:")
print(df[["Machine", "Vibration_mm_s", "Condition"]])

```

Sample Input and Output :

Input: The machine data are predefined in the Python dictionary in the program. **No keyboard input** is required during execution.

Machine	Speed_RPM	Temperature_C	Vibration_mm_s	Power_kW
M1	1200	65	2.1	5.2
M2	1500	72	2.8	6.5
M3	1800	85	4.5	8.0
M4	1400	68	2.4	5.8
M5	1600	78	3.2	7.1

Expected Output:

Machine Data:

	Machine	Speed_RPM	Temperature_C	Vibration_mm_s	Power_kW
0	M1	1200	65	2.1	5.2
1	M2	1500	72	2.8	6.5
2	M3	1800	85	4.5	8.0
3	M4	1400	68	2.4	5.8
4	M5	1600	78	3.2	7.1

Average Temperature: 71.6 °C

Average Vibration: 3.0 mm/s

Average Power: 6.52 kW

Maximum Vibration: 4.5 mm/s

Machine with Maximum Vibration: M3

Machine Condition:

	Machine	Vibration_mm_s	Condition
0	M1	2.1	Normal
1	M2	2.8	Normal
2	M3	4.5	Needs Inspection
3	M4	2.4	Normal
4	M5	3.2	Needs Inspection

Result : The Python program was written and executed successfully for analyzing machine operating data using NumPy and Pandas, calculating average values, identifying the maximum vibration and classifying machine conditions.